

Effects of Soldering Process Parameters on Lead-Free Solder Joint Shear Strength

IE 533 Final Project Report

Background

The electronics manufacturing industry underwent a fundamental shift in soldering materials following the adoption of the European Union's Restriction of Hazardous Substances (RoHS) directive, which took effect on July 1, 2006, and effectively banned the use of lead-based solder alloys in consumer electronics equipment [3]. The most widely adopted lead-free replacement is SAC305, which is an alloy composed of 96.5% tin (Sn), 3.0% silver (Ag), and 0.5% copper (Cu) [1]. It provides adequate wettability and mechanical performance in electronics assembly. SAC305 has a solidus temperature of 217°C and a liquidus temperature of approximately 220°C [1]. Hence, SAC305 soldering requires peak temperatures in the range of 250-260°C to ensure complete solder melting, wetting, and intermetallic compound (IMC) formation [4].

Solder joint shear strength is defined as the force per unit area required to fracture a solder joint under lateral loading. It is one of the most critical mechanical quality metrics in electronics assembly. Insufficient shear strength leaves joints susceptible to premature failure under vibration, thermal cycling, and handling stresses, which is a primary failure mode in automotive, aerospace, and consumer electronics [2]. Hence, proper process optimization to maximize joint shear strength is a central objective in electronics manufacturing process engineering.

This experiment addresses the following: Which soldering process parameters from soldering temperature, solder paste volume, and post-solder cooling rate have statistically significant effects on the shear strength of SAC305 solder joints, and do any of these factors interact?

Outcome Variable and Measurement

The outcome variable is **shear strength, measured in N/mm²**. Shear strength is measured via a lateral push test using a calibrated bond tester and yields a continuous, quantitative response suitable for ANOVA.

Factors and Levels

Three design factors were selected based on their relevance in lead-free soldering literature:

Table 1: Factors, their levels, and reasoning

Factor	Low-level	High-level	Reasoning
Soldering Temperature (A)	230°C	260°C	Spans the practical SAC305 reflow window. 230°C provides a margin above the 217°C solidus, while 260°C is the recommended upper limit to avoid thermal damage.
Soldering Volume (B)	0.5 mg	1.0 mg	Represents minimal vs. standard SMD pad paste deposit. Insufficient volume risks incomplete joints, while excess volume risks solder bridging between adjacent pads.

Cooling Rate (C)	1°C/s	4°C/s	Represents natural convective cooling (slow) vs. forced air quenching (fast), spanning the range achievable in a standard reflow oven cooling zone [4].
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Experimental Design

A **2³ full factorial design** was selected, consisting of three factors each at two levels as highlighted in Table 1, producing 8 unique treatment combinations. The design is fully crossed, hence every possible combination of factor levels is tested. This is a **fixed-effects model**, as all factor levels were chosen deliberately based on process knowledge rather than randomly sampled from a larger population. The design supports estimation of all seven effects: three main effects (A, B, C), three two-way interactions (AB, AC, BC), and the three-way interaction (ABC).

Two replicates per combination were used, yielding $8 \times 2 = 16$ total runs. The resulting degrees of freedom for pure error are $df_E = 8 \times (2 - 1) = 8$, meeting the accepted minimum threshold for reliable F-test estimation in factorial designs. Three replicates were considered but deemed unnecessary as a third replicate would increase experimental effort by 50% with only marginal power gain.

The following nuisance variables were held constant throughout the experiment to reduce experimental error and increase the sensitivity of F-tests:

- Solder alloy: SAC305 exclusively (96.5% Sn, 3.0% Ag, 0.5% Cu) for all runs
- PCB substrate and pad geometry: same board and copper pad dimensions across all runs
- Preheat temperature: fixed at 120°C (held constant as a nuisance factor since uncontrolled preheat variation could confound the effect of soldering temperature)
- Ambient conditions: constant room temperature and humidity throughout
- Equipment: same reflow oven and forced-air cooling apparatus for all 16 runs

The run order of all 16 runs was fully randomized using Minitab's built-in run order randomization at design creation. This ensures time-dependent nuisance effects are spread uniformly across treatment cells rather than confounded with specific factor levels.

Blocking was not done, as all 16 runs were conducted on a single soldering station with no natural grouping variable. Adding blocks would reduce error df without reducing error variance.

A power analysis was conducted using the following assumptions: significance level $\alpha = 0.05$, minimum detectable effect $\delta = 5 \text{ N/mm}^2$ (corresponding to the anticipated magnitude of the Cooling Rate main effect), within-cell standard deviation $\sigma = 3 \text{ N/mm}^2$ (reflecting expected within-cell variability in solder joint mechanical testing), and 2 replicates. The analysis yielded an **actual power of about 0.83**, exceeding the conventional 0.80 minimum threshold. This confirms that 16 total runs provide sufficient sensitivity to detect a practically meaningful 5 N/mm² difference in shear strength.

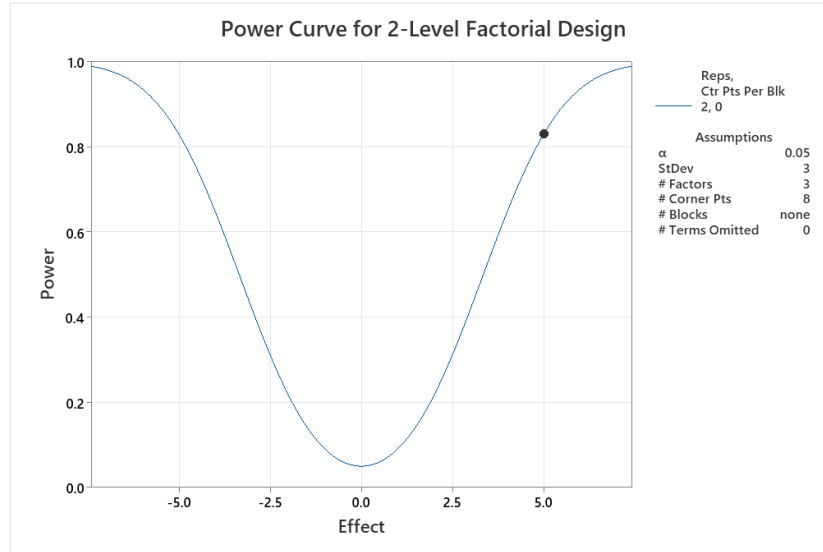


Figure 1: Power curve for 2^3 full factorial design

Data

The response data was generated by simulation, as conducting physical soldering experiments is not within the scope of this project. The following data generating function (DGF) was used:

$$\text{Shear Strength} = 35 + 4 \cdot \text{Soldering Temperature} (260) + 3 \cdot \text{Soldering Volume} (1.0) + 5 \cdot \text{Cooling Rate} (4) + 3 \cdot \text{Soldering Temperature} (260) \cdot \text{Cooling Rate} (4) + \varepsilon$$

where $\varepsilon \sim N(0, 3)$

The baseline of 35 N/mm² reflects realistic SAC305 conditions, and coefficient magnitudes reflect relative effects from the literature. The Temp $\times$ Cooling Rate interaction (+3 N/mm²) reflects the hypothesis that high-temperature IMC benefits are amplified by rapid cooling, and $\varepsilon \sim N(0, 3)$ represents measurement variability [4].

Numerical Summary

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Shear Strength	16	0	42.6203	1.32505	5.30019	33.5918	38.6852	42.4616	45.8178	52.2389

Figure 1: Numerical Summary for Shear Strength

Mean shear strength is 42.62 N/mm² (StDev = 5.30), ranging from 33.59 to 52.24 N/mm². The near-equal mean and median (42.46) confirm symmetry consistent with the normal noise term.

Data Analysis

A general linear model (GLM) was fit to the full factorial design and ANOVA was used to test main effects and all interactions at $\alpha = 0.05$. The following null hypotheses were tested by fitting a GLM to the full factorial design: $H_0: \tau_A = 0$; $H_0: \tau_B = 0$; $H_0: \tau_C = 0$; $H_0: (\tau\beta)_{ij} = 0$ for all i, j .

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	7	365.018	52.145	7.40	0.006
Linear	3	354.924	118.308	16.79	0.001
Soldering Temperature	1	190.463	190.463	27.03	0.001
Soldering Volume	1	45.257	45.257	6.42	0.035
Cooling Rate	1	119.204	119.204	16.92	0.003
2-Way Interactions	3	8.195	2.732	0.39	0.765
Soldering Temperature*Soldering Volume	1	1.774	1.774	0.25	0.629
Soldering Temperature*Cooling Rate	1	0.792	0.792	0.11	0.746
Soldering Volume*Cooling Rate	1	5.629	5.629	0.80	0.397
3-Way Interactions	1	1.899	1.899	0.27	0.618
Soldering Temperature*Soldering Volume*Cooling Rate	1	1.899	1.899	0.27	0.618
Error	8	56.362	7.045		
Total	15	421.380			

Figure 2: Analysis for Variance table for Shear Strength

The overall model is statistically significant ($p = 0.006$), confirming that at least one factor meaningfully affects shear strength. Linear main effects collectively account for 354.924 of the 365.018 total model SS. All three main effects are significant, and all interaction terms are not.

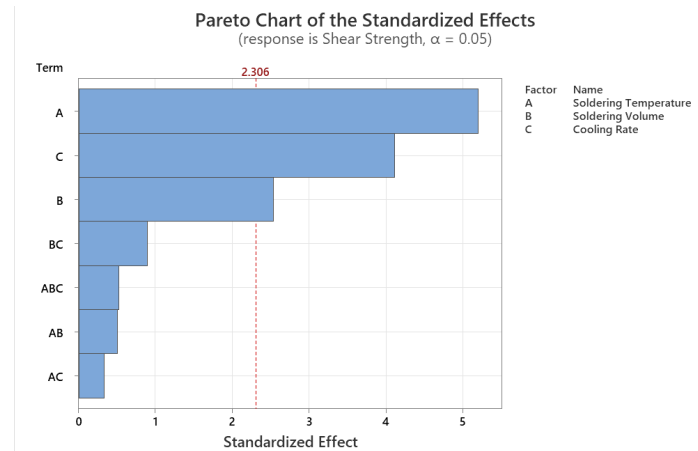


Figure 3: Pareto chart of standardized effects for Shear Strength

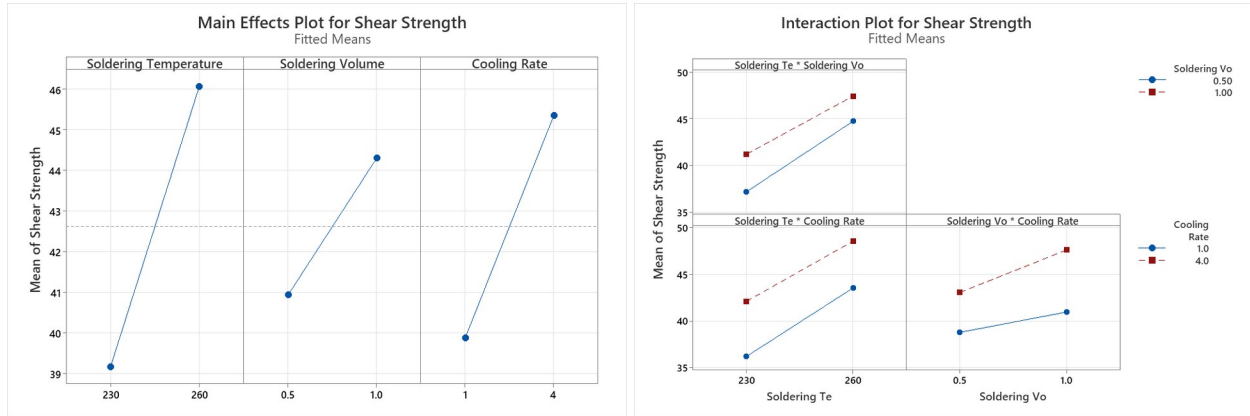
The Pareto chart in Figure 3 visually confirms the ANOVA conclusions. Factors A (Soldering Temperature, standardized effect ≈ 5.2), C (Cooling Rate, ≈ 4.1), and B (Soldering Volume, ≈ 2.5) all exceed the $t = 2.306$ reference line. All interaction terms fall well below the threshold.

Coded Coefficients

Term	Effect	Coef	SE Coef	T-Value	P-Value	VIF
Constant		42.620	0.664	64.23	0.000	
Soldering Temperature		6.900	3.450	0.664	5.20	0.001
Soldering Volume		3.364	1.682	0.664	2.53	0.035
Cooling Rate		5.459	2.730	0.664	4.11	0.003
Soldering Temperature*Soldering Volume		-0.666	-0.333	0.664	-0.50	0.629
Soldering Temperature*Cooling Rate		-0.445	-0.222	0.664	-0.34	0.746
Soldering Volume*Cooling Rate		1.186	0.593	0.664	0.89	0.397
Soldering Temperature*Soldering Volume*Cooling Rate		0.689	0.344	0.664	0.52	0.618

Figure 4: Coded Coefficients table for Shear Strength

The estimated effects in coded units are: Soldering Temperature (+6.90 N/mm²), Cooling Rate (+5.459 N/mm²), and Soldering Volume (+3.364 N/mm²). The intercept of 42.620 N/mm² equals the overall mean, consistent with the descriptive statistics. All VIF values equal 1.00, confirming design orthogonality and zero multicollinearity among factor estimates.



Figures 5 & 6: Main Effects and Interaction Plot for Shear Strength respectively

The main effects plot (Figure 5) shows all three factors positively influence shear strength. Moving from low to high, Soldering Temperature produces the steepest increase, followed by Cooling Rate, then Soldering Volume. All three points deviate substantially from the grand mean reference line. The interaction plot (Figure 6) presents all three pairwise factor combinations. Lines across all panels are approximately parallel, consistent with no statistically significant interactions.

A notable and interesting feature of this simulated dataset is the discrepancy between the DGF and the statistical results. The DGF includes a Soldering Temperature $\times$ Cooling Rate interaction of +3 N/mm², however, the ANOVA revealed this interaction was not statistically significant ($p = 0.746$), and it does not cross the reference line in the Pareto chart. This illustrates the fundamental principle that statistical power is effect-size dependent. Here, the lines remain almost parallel, reflecting the design's insufficient power to detect this smaller interaction at 2 replicates.

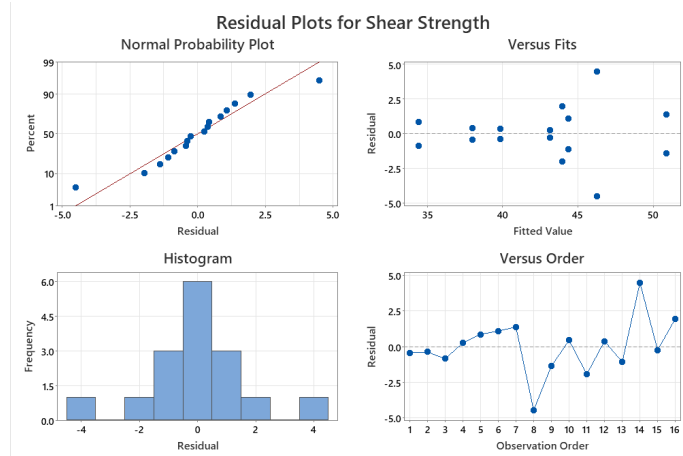


Figure 7: Four-in-one residual diagnostic plots for Shear Strength

The residual plots confirm normality, constant variance, and no time-based trend, satisfying all GLM assumptions. The two observations (observations 8 and 14) flagged reflect large noise draws from the same treatment cell and are consistent with the $N(0,3)$ distribution.

Key Findings

Soldering Temperature had the largest effect ($+6.9 \text{ N/mm}^2$), followed by Cooling Rate ($+5.5 \text{ N/mm}^2$) and Soldering Volume ($+3.4 \text{ N/mm}^2$), consistent with literature on IMC formation and grain microstructure. No interaction effects were statistically significant, despite a Soldering Temperature $\times$ Cooling Rate interaction being built into the DGF, illustrating the power limitations of small-replicate designs for detecting moderate interactions. The optimal factor combination of high temperature (260°C), large volume (1.0 mg), and fast cooling (4°C/s) provides actionable guidance for reflow profile optimization in lead-free SMT assembly.

References

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